

NutriWise AI - Replit Integration Specification

Purpose

This document serves as the master implementation guide for integrating the existing VitalGuide/NutriWise website with five CrewAI agents. Preserve the existing project and extend it; do not rewrite.

Architecture

Frontend: React. Backend: Express/Node.js. Database: PostgreSQL (Drizzle). Authentication: Clerk. AI: CrewAI Studio. Backend acts as orchestrator; CrewAI performs reasoning.

Agents

Agent1 Medical Report Intelligence; Agent2 Conversational Health Assessment; Agent3 Intelligent Care Planner; Agent4 Emergency Navigator; Agent5 Nutrition RAG.

Integration Rules

Never call CrewAI directly from React. All requests pass through an AgentRouter. Persist structured outputs. AI never writes directly to DB or sends emails/SMS.

Backend Modules

Create ai/AgentRouter.ts, CrewAIClient.ts, ResponseParser.ts, AIWorkflowManager.ts, CrewAIMapper.ts. Add services: NotificationService, EmailService, SMSService, EmergencyService.

Suggested APIs

POST /api/ai/checkup, /report, /planner, /nutrition, /emergency, /chat; GET /api/dashboard, /profile.

Workflow

Medical report -> Agent1 -> Agent2 (if needed) -> Agent5 (nutrition) -> Agent3 (care plan) -> Agent4 (if emergency). Chat routes intelligently based on intent.

Frontend

Enhance existing Dashboard, Checkup, Planner, Education/Nutrition, Profile, Emergency pages. Add upload, streaming chat, health cards, task lists, emergency banner, nutrition evidence.

Database

Extend existing schema with AI runs, assessments, care plans, nutrition plans, emergency events, notifications, health scores, compliance reports, uploaded reports.

Security

Protect APIs with Clerk, validate uploads, never expose secrets, sanitize inputs, enforce file limits.

Performance

Cache profile, plans, reports, conversation summaries. Only rerun agents when relevant inputs change.

Future

Design plug-in architecture for Mental Health, Fitness, Vision, Wearables, Medication Interaction, Lab Trend agents.

Implementation Notes: Preserve existing UI and routing. Build orchestration layer, not AI logic in the frontend. Backend executes notifications/emails based on agent payloads. Maintain modularity and extensibility.